Performed, Not Observed
Text as Causal Substrate

Joseph A. Wecker

Independent Researcher, Vis Veritatis (v2.io), Lehi, USA joseph.wecker@v2.io

Abstract

For a decade one sentence has carried the dismissal of language models as causal reasoners: they are trained on text, and text is observational data, so they are bounded at Pearl's associational rung. The load-bearing clause is a category error. Cooperative declarative text is not a table of co-occurrences; it is a record of *performed* causal commitments — "the frost killed the crop" asserts an intervention, not a correlation. Under three named postulates, the information distinguishing a causal claim from its directional reverse is, by the Causal Hierarchy Theorem, not recoverable from any Level-1 distribution over the event variables alone (a lower-bound conditional result). This exposes a seam the field measures across without naming: *encoded* content (what a representation holds) versus *deployed* content (what a forward pass emits); which side an evaluation lives on fixes what it can ever certify. We build the substrate-side instrument — formal semantics as oracle, no annotation — that does not merely detect a directional signal but **decomposes** it into the route the theory licenses (a token bag cannot reproduce it) and the route a Level-1 corpus summary could, with a symmetric control for a world-knowledge prior; it runs on a 12-model panel and on decoder hidden states themselves. Companion work proves behavior cannot even *lower-bound* the encoded-presence precondition the instrument discharges: the relationship is a modality-switch handoff at the seam, not a duality. Run across the panel, the instrument delivers exactly the kind of verdict behavior provably cannot reach: on the cleanest test the non-reducible route is rare with no capacity trend, and the panel's largest model's apparently strong real-world result is a direction prior. The theorem dissolves the confident dismissal *a priori*; the empirics decline the confident attribution; the symmetric cut is enacted, not argued. We give the theorem, the instrument, and the prescription that put the question on honest footing.

1 Introduction

A sentence that has done too much work. For a decade one sentence has carried the dismissal of language models as causal reasoners: *they are trained on text; text is observational data; so they are bounded at Pearl's associational rung* (Pearl 2018). The empirical record assembled against and around that claim (Zecevic et al. 2023; Jin et al. 2023; Chi et al. 2024) is, taken on its own terms, hard to reconcile with the dismissal and with itself — a literature shaped like a contradiction. We do not adjudicate it. We examine the sentence.

The category error. The load-bearing clause is the second: *text is observational data.* A stream of tokens, considered as raw occurrence statistics, is observational. But the content those tokens convey under ordinary cooperative discourse is not. When a writer commits "the frost killed the crop," the conventional semantics of *killed* asserts an interventional claim — a counterfactual dependence, a what-would-have-happened — not a report that frost and crop loss were correlated. Text is a record of **performed causal commitments**. Under three named postulates from linguistic theory — that causal markers carry conventional content at their Pearl level (SLC), that a competent speaker using them in declarative context is committed to that content (SC), and that text-level content composes from marker-level commitments (CS) — the marker structure of cooperative declarative text carries interventional and counterfactual commitments by convention, and the information that distinguishes a causal claim from its directional reverse is, by the Causal Hierarchy Theorem (Bareinboim et al. 2022), *not* recoverable from any Level-1 distribution over the event variables alone (§3, Theorem 1; a lower-bound conditional result, audited at the load-bearing non-reduction step). The decade-old dismissal mis-describes its own input: it treats performed causal commitment as observed correlation, and concludes a bound that the mis-description, not the models, produces.

The seam the field has measured across without naming. Correcting the description exposes a distinction the literature has been measuring across blindly. *Encoded* content is what survives in a model's representation, present whether or not any output invokes it. *Deployed* content is what the forward pass actually emits. The two come apart, and **which side an evaluation lives on fixes what it can ever certify.** This is not a slogan; it is the organizing fact of the paper. The substrate-side question — *is the mechanism-distinguishing causal content present and non-reducible in the representation?* — is answerable. The deployment-side question — *does the model use it?* — is, for the dominant architecture, structurally out of reach of behavioral evidence (companion work; §9). A behavioral failure is therefore not evidence of an encoding absence, and a behavioral success is not evidence of genuine encoding; the field's recurring confusion is, in large part, a single unnamed seam.

The instrument, and the discipline that keeps it honest. We build the substrate-side instrument: a class of probes that use formal semantics itself as oracle — no annotation, no crowd-sourcing — and that do not merely *detect* a directional signal but **decompose** it. One route is the one the theory licenses (the geometric distinction survives only with word order intact; a Level-1 token bag cannot reproduce it). The other a sufficiently rich Level-1 corpus summary could in principle reproduce (vocabulary co-occurrence; Reichenbachian inheritance, not "shallow pattern-matching"). A symmetric control strips a world-knowledge prior that otherwise rides along. The decomposition is the discipline: only the first route tests the theorem's non-reducibility property, and the paper never lets a result of the second kind borrow the first's authority. The same instrument, unchanged, reads a non-causal directional domain (evidential register, §7), so it is a substrate inventory tool, not a one-claim probe — and it runs not only on embedding models but on the pooled hidden states of decoder language models themselves: the seam is live on the very architecture the behavioral debate is conducted on.

What is proven, and what is reported. Two results carry the paper's weight. **Theorem 1** (§3), proven here under three named postulates: the encoded content is present and non-reducible in the text. **The certification statement** of §9, derived from the companion no-go under named premises: this paper's relationship to that no-go is a *modality-switch hand-off at the encoded/deployed seam*, in which the substrate diagnostic discharges exactly the encoded-presence precondition behavioral evidence cannot even lower-bound, and the same seam is precisely why presence licenses no claim about deployed use. The empirical panel is the instrument exercised. On the cleanest test of the theory-licensed route — a vocabulary-identical contrast a token bag cannot reproduce — the non-reducible signal appears in only a minority of the twelve models with no capacity trend, and a symmetric control shows the apparently strong real-world results, conspicuously on the largest panel model, are substantially a world-knowledge-direction prior, a pattern that reproduces on independently-generated test data (§5–§6). The decomposition converts "language models encode causal structure" from an unfalsifiable slogan into a measured claim that is mostly the vocabulary-recoverable route — a result behavioral evaluation provably cannot produce, because it cannot reach this side of the seam at all.

Stakes. The practical prescription is adoptable now: evaluate encoded content and deployment fidelity separately; treat a behavioral failure as silent on encoding; place representation-level audit in the safety toolkit beside, not instead of, behavioral testing. The societal stake is sharper. The dismissal we reframe is routinely used to ground *confident* denials in the welfare and capability debate — "they only pattern-match" deployed as a premise rather than earned as a finding. We refute one specific empirical ground for that confidence at the representation level. The discipline that forbids projecting an upper bound from below cuts symmetrically, against confident attribution and confident dismissal alike. The contribution to the debate is that symmetry.

Roadmap. §2 situates the work among representational and behavioral probes. §3 develops the theoretical anchor (the three postulates; Theorem 1 as an audited lower-bound conditional result; the C1/C2/C3/C4 mechanism framework). §4 develops the diagnostic as a methodology. §5–§6 exercise it across the panel and report the mechanism decomposition and its breadth. §7 establishes class-generality on evidential register. §8 is the honest limitations ledger. §9 states the encoded/deployed-seam certification relationship to the companion no-go. §10 concludes.

2 Related Work

Behavioral causal-reasoning benchmarks. A line of work prompts LLMs with causal-reasoning tasks and grades generated output: Causal Parrots (Zecevic et al. 2023), CLadder (Jin et al. 2023), CausalProbe (Chi et al. 2024), and the counterfactual-rule benchmarks. These measure a generative channel's deployed behavior. Our diagnostic is substrate-side and complementary, not competitive: it measures what a frozen representation makes available, not what a channel does with it (§3, §9). The two paradigms together would pin down whether a behavioral failure is encoding-side or deployment-side; neither alone can.

Representational probes for causal/relational content. The closest framing-neighbor is Willig et al. (2023), probing LLM embeddings for ConceptNet-derived causal-fact clustering — same group as Causal Parrots. The differences are the contribution: our oracle is Pearl semantics itself, not a knowledge base; our target is Pearl-Level-2/3 *specificity* and its CHT-non-reducibility, not causal-vs-non-causal classification; our probe construction is Pearl-entailment geometry, not KB clustering; and we *decompose* the detected signal by mechanism rather than reporting detection. Lepori et al. (2026)'s modal difference vectors from decoder activations on event-plausibility categories is the second neighbor: decoder activations during generation versus frozen representations, event-plausibility versus Pearl-Level-3 counterfactual modal strength, human-rating-validated versus semantics-as-oracle. GaussCSE-style directed sentence embeddings (Yoda et al. 2024) encode NLI-entailment direction; Pearl-direction-with-counterfactual-entailment is a different target than NLI-entailment-direction and we say so explicitly.

Pearl-hierarchy foundations and information-theoretic separations. The hierarchy and its non-collapse are developed in Pearl (2009) and Bareinboim et al. (2022); we use the hierarchy as the layer-naming device and the non-reduction theorem as the formal anchor for the encoded-content lower bound (§3), and do not contribute to the hierarchy's foundations. Quantitative information-cost separations between the Pearl levels (the Causal Description Gap line of work) are adjacent and cited;

our theoretical-spike layer engages them separately and they are not load-bearing here. Concurrent work (Joshi et al. 2026) independently argues the Pearl hierarchy is the right diagnostic vocabulary for interpretability claims, classifying observational, interventional, and counterfactual evidence by the rung it can support — a typing the present diagnostic respects in distinguishing C1 (CHT-non-reducible, the Theorem-1 route) from C2 (Reichenbachian, Level-1-recoverable).

Companion methodology-critique work. Companion work proves an architectural floor on what behavioral evaluation of decoder-class LLM agents can certify about causal-access mechanism. Its relationship to this paper is stated precisely in §9: a modality-switch handoff at the encoded/deployed seam — behavioral evaluation provably cannot lower-bound the encoded-presence precondition this diagnostic discharges — not a duality and not the same measurement.

Positioning. Under the narrow framing — substrate-side structural probes on frozen representations, formal-semantics-as-oracle, mechanism-decomposing, capacity-discriminative — the prior-art audit assessed the core conjunction as novel; under a broad framing ("first probe of causal content in LLMs") it would be rejected on prior art, which we do not claim. We pre-empt the two closest neighbors in this section by explicit differentiation and report an arXiv concurrent-work sweep at submission time.

3 Theoretical Anchor

Setup

We work in the Pearl–Bareinboim causal hierarchy over a structural causal model $M = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ with endogenous event variables $\mathbf{V}$ and layers $\mathcal{L}_1 \subset \mathcal{L}_2 \subset \mathcal{L}_3$ (observational, interventional, counterfactual). The Causal Hierarchy Theorem (CHT, Bareinboim et al. 2022) establishes that for almost every model the $\mathcal{L}_2$ and $\mathcal{L}_3$ quantities are not identifiable from $\mathcal{L}_1$ alone. "Pearl-Level-2 commitment" denotes propositional content whose truth-conditions are stated at $\mathcal{L}_2$ — not the use of any particular formal apparatus.

Three postulates

The result is conditional on three postulates about the relation between language use and event-level structure. They are stated as postulates because they are facts about how communication systems develop, not derivable from non-linguistic first principles; each is defended in the linguistics literature.

(SLC) Standard Linguistic Convention. Explicit causal-marker constructions carry conventional semantic content at their Pearl level: a speaker who accepts the marked utterance while denying the corresponding $\mathcal{L}_2/\mathcal{L}_3$ quantity contradicts the language as conventionally used. A claim about conventional semantics, not speaker introspection.

(SC) Speaker Commitment. A competent speaker using a causal marker in a declarative, non-ironic, non-quoted context is committed to its conventional content. (SC) bridges *marker present in text* to *content asserted*. We adopt the weak, corpus-scale form: it need not hold sentence-by-sentence, only on average across the genres a training corpus draws from.

(CS) Compositional Structure. Text-level content composes from marker-level commitments by standard compositional-semantics rules; the text's causal content is assembled from the markers and their relata, not an unrelated emergent property.

(SLC) is the load-bearing semantic claim; (SC) is load-bearing *historically*, in the production of the training corpus, not at probe time (see §4 and §8); (CS) lets the marked content be read off compositionally.

Theorem 1 (lower-bound conditional result)

> **Theorem 1.** Under (SLC), (SC), (CS), for any natural-language corpus containing explicit causal-marker constructions used cooperatively, the marker structure encodes propositional content whose truth-conditions lie at $\mathcal{L}_2$ and that is **not derivable from the joint distribution over the event variables V alone**.

It is a **lower bound**: it asserts presence and non-reducibility of the asserted content, not its truth, completeness, or fidelity, and *not* that any learned system recovers or deploys it. The non-reduction is asserted at the **event-variable level** — not derivable from $P(\mathbf{V})$ — and not at the text-token level, where the marker tokens trivially distinguish the cases.

Lemma 1 (the kettle pair). T_B = "the water boiled, *causing* the kettle to whistle" and T_D = "the kettle whistled, *causing* the water to boil" are committed to identical $\mathcal{L}_1$ content over $\{\text{boil}, \text{whistle}\}$ (both events occurred) but opposite $\mathcal{L}_2$ content. Structural models $\mathcal{M}_B$: boil $\rightarrow$ whistle and $\mathcal{M}_D$: whistle $\rightarrow$ boil can be made observationally equivalent by choice of exogenous distributions; by CHT no $\mathcal{L}_1$ analysis of the event-variable joint distribution separates them. The distinguishing information is carried entirely by marker structure. The audit verifies the construction generalizes to any event-pair admitting both causal directions, and scopes out three edge cases (deterministic, cyclic, self-referential).

What Theorem 1 does not establish: that any commitment is *true* (recovery is not verification); that any substrate recovers it *faithfully* (the substrate/channel question, §9); that recovery is unique or complete (explicit-marker content only — a lower bound).

The C1/C2/C3/C4 mechanism framework

Theorem 1 is silent on *how* a learned representation would carry the content. The empirical work separates four candidate routes.

- **C1 — syntactic-position route.** The geometric distinction lives in vocabulary-identical minimal pairs differing only in argument position; nulls under token-shuffle. This is the route Theorem 1 most cleanly licenses and the empirical analogue of CHT-non-reducibility: a Level-1 token-bag cannot reproduce it.

- **C2 — Reichenbachian inheritance.** Lexical-pattern statistics tracking real-world causal frequencies (Reichenbach's common-cause principle (Reichenbach 1956) at word-co-occurrence scale). C2 survives token-shuffle and is *not* CHT-non-reducible: a sufficiently rich Level-1 vocabulary distribution can carry it. Framed positively, not pejoratively: C2 is the substrate inheriting source-causal structure via distributional co-occurrence, not "shallow pattern-matching."
- **C3 — ICM time-asymmetry.** Independent-causal-mechanism direction asymmetry; directional asymmetry derives, a quantitative bound does not (precisely-specified open target).
- **C4 — Causal-IB consequence.** IB-optimal compression preserves causal structure proportional to predictive value (existing framework machinery instantiated on linguistic data; a plausibility bridge, flagged in §8).

The operationally load-bearing split is **C1 vs C2**, decided by the joint test (vocabulary-identity $\times$ token-shuffle behavior). Only **C1-dominant** classifications test the Theorem-1 non-reducibility property; C2-dominant and MIXED classifications are evidence of Pearl-content-*aligned* signal whose mechanism a Level-1 corpus summary could in principle reproduce (this distinction is load-bearing for §8–§9 and is the honest scope boundary on the theory-to-empirics bridge). The **H_D3** symmetric control further separates genuine C1 syntactic-position encoding from a world-knowledge-direction prior that rides along with canonical-direction event pairs.

Relation to behavioral certification (forward pointer)

Theorem 1 is a claim about what is present in the substrate that any channel reads from; it is not a claim about deployment. §9 makes the cross-paper relationship precise: behavioral evaluation provably cannot even lower-bound the encoded-presence precondition this diagnostic discharges, and the encoded/deployed seam is exactly why presence does not upgrade to deployed use. The connection is a modality-switch handoff (proven under named premises; companion work), not a duality.

4 The Diagnostic Class

What the diagnostic is

A class of probes that test whether a **frozen** pretrained representation structurally encodes content at a specified level of Pearl's hierarchy (or, more generally, any directionally- or categorially-structured semantic property with stable conventions). Each probe uses **Pearl-semantic natural readings under standard interventionist assumptions** (Pearl 2009; Lewis 1973) as oracle — no human ratings, no crowd-sourcing, no expert annotation. The theory supplies the predicted geometric relations; the construction is a minimal pair plus the relations its semantics dictates among embedding similarities.

Three properties distinguish it from behavioral causal-reasoning benchmarks: it operates on a frozen forward pass (no prompting, decoding, or generation); its ground truth is the formal semantics itself, not annotation; and it isolates *encoded* content from *deployed* behavior — the distinction §3's Theorem 1 makes precise and §9 makes load-bearing.

The joint test that decomposes mechanism

Detection alone does not separate the route Theorem 1 licenses (C1, CHT-non-reducible) from the route a Level-1 corpus summary could reproduce (C2, Reichenbachian vocabulary co-occurrence). The diagnostic's defining move is a **joint test** over two axes:

- **Vocabulary-identity of the contrast.** A vocabulary-identical contrast (same tokens, only argument position or grammatical voice differs) isolates C1; a vocabulary-different contrast (markers or events differ) admits both C1 and C2.
- **Token-shuffle behavior.** Multi-shuffle (the bag of tokens preserved, word order destroyed, averaged over many independent shuffles to marginalize any single-permutation out-of-distribution artifact). A signal that nulls under shuffle on a vocabulary-identical contrast is C1; one that survives is C2.

The per-test classification (C1-dominant / MIXED / C2-dominant / the named ANOMALOUS subcategories / ORIG-NULL) is computed from the original-order and multi-shuffle effect with a **paired bootstrap** ratio CI over pair indices (preserving within-pair covariance) and reported with **two** uncertainty intervals: a noise-reduced one (mean-shuffle-per-pair) and a full-uncertainty one (joint resample of pair and shuffle indices). An **H_D3** symmetric control — does the reverse causal claim prefer the reverse counterfactual as strongly as the forward claim prefers the forward one? — separates genuine syntactic-position encoding from a world-knowledge-direction prior on canonical event pairs.

Probe instruments

The instrument family comprises: a C1/C2 mechanism-classification matrix (E14) over the canonical alignment probes E1-E13; a real-world H_D1 + H_D3 split by association strength (E15); a five-level lexical-disruption gradient L1$\rightarrow$L5 that decomposes the contribution of world-knowledge vocabulary, of Level-2/3 and Level-1 marker vocabulary, of entity recurrence, and — at L5, the construction-family scrub — of register vocabulary, against a structural residual (E16); a substrate-vs-channel cloze test with multi-shuffle and direction-axis-permutation nulls (E17); a pair-difference linear probe with held-out-pair CV, label-swap and pair-permutation nulls, and an OOD-marginalized bag control (E18); and a cross-substrate validation on test data generated by independent LLM sources (E19). All emit JSON with seeds, model IDs, Cohen's d, Bonferroni-corrected p-values, and a `design_version` provenance field.

Properties that make it a class, not a single probe

- **Substrate-agnostic.** Any system producing a sentence representation — encoder, decoder pooled hidden states, MoE. The decoder-pooled-hidden-state route makes the substrate/channel distinction empirically live on the same model class behavioral benchmarks evaluate.

- **Annotation-free.** The theory is the oracle.
- **Self-correcting.** The diagnostic surfaces its own confounds and distinguishes a surface-form artifact from a genuine encoding gap via a cleaner-template variant — a methodological capability, not only a result.
- **Capacity- and mechanism-discriminative.** It does not pass/fail a model; it reports *by what route*, at what capacity, with what confound structure.
- **Content-general.** The same architecture applies to a non-causal directional/categorial domain (evidential register; §7), which is what makes it a frozen-substrate inventory tool rather than a Pearl-specific probe.

What the diagnostic does not do

It measures the substrate, not the channel: a C1-dominant result certifies that mechanism-distinguishing content is present and CHT-non-reducible in the representation; it is silent on whether a generative deployment uses it. That boundary is not a limitation to be engineered away — §9 shows it is exactly the seam at which this method and the companion behavioral no-go compose.

5 Empirical Findings

Panel and protocol

Twelve frozen pretrained models spanning a ˜330× capacity range (23M–7.6B parameters), with four within-family pairs for capacity-versus-architecture disentangling, plus decoder pooled hidden states for the substrate/channel route. Each probe carries an identity sanity check, paired one-tailed Wilcoxon, paired-bootstrap CIs (noise-reduced and full-uncertainty), a multi-shuffle noise floor, Cohen's d, Bonferroni correction across the test battery, and JSON provenance (seeds, model IDs, `design_version`). The oracle is Pearl-semantic natural readings under standard interventionist assumptions; there is no annotation.

The cleanest test of the theory-licensed route is the decisive one

The vocabulary-identical contrast (E2: active-vs-passive voice, same tokens, only argument position differs) is the only test that isolates the CHT-non-reducible route a token bag cannot reproduce — the route Theorem 1 licenses. Across the twelve models it classifies **C1-dominant in 3, C2-dominant in 1, MIXED in 1, ORIG-NULL in 4, and ANOMALOUS-sign-flip in 3**. The non-reducible route is the minority outcome on the cleanest test, with no capacity monotonicity (Spearman $\rho \approx -0.08$ on the shuffle/orig ratio, not significant; the smallest model and the largest both land C1-dominant while the mid-capacity band is null or anomalous).

The vocabulary-different contrasts look stronger but, by construction, admit the vocabulary-recoverable route and so do not test the Theorem-1 property: E1 (cause-vs-temporal) is C1-dominant in only 3 of 12 (3 C2-dominant, 3 ORIG-NULL, 3 anomalous); E3 (cross-marker) is 4 C1 / 5 MIXED / 3 C2; E10 (modal strength — the V1 "30/30-perfect" headline) decomposes to 2 C1 / 4 MIXED / 1 C2 / 5 ANOMALOUS-amplify, i.e. the V1 result was substantially the bag route once the multi-shuffle control is applied.

The world-knowledge-direction prior is real and hits the largest panel model

On real-world entities the symmetric control (E15: does the reverse causal claim prefer the reverse counterfactual as strongly as the forward claim prefers the forward one?) splits the panel as **BIDIRECTIONAL in 4, ASYMMETRIC — the direction-prior-confound signature — in 5, REVERSE-only in 1, NULL in 2**. The genuine bidirectional set is fewer than half the panel and is not capacity-ordered. qwen3-embedding, the largest model and the V1 flagship for "Pearl-Level-2 specificity," is **ASYMMETRIC**: a large forward effect (H_D1 $= +0.025$, CI $[+0.015, +0.035]$) with a symmetric effect whose CI crosses zero (H_D3 $= +0.009$, CI $[-0.001, +0.018]$) — the world-knowledge-direction-prior signature, not clean syntactic-position encoding. The pattern reproduces on test data generated by an independent model (E19 codex_c1: 2 BIDIRECTIONAL, 3 ASYMMETRIC, 7 NULL/REVERSE across 12), if anything sparser than on the hand-constructed pairs.

The pair-difference probe (E18) does not rescue the strong reading on the flagship: qwen3 held-out-pair CV is 0.45, at chance, and its OOD-marginalized bag CV of 1.000 is the variance-shrinkage artifact §8 flags, not bag-recoverability. (The full E16 lexical-disruption gradient and its L4–L5 register-vocabulary decomposition are reported in §6; the E17 substrate-vs-channel cloze nulls and the E18 per-model panel in their respective tables.)

Summary

Measured cleanly and decomposed, "language models encode causal structure" is mostly the vocabulary-recoverable route; the CHT-non-reducible route is rare and not capacity-monotone; and the strongest-looking real-world evidence is a world-knowledge-direction prior the symmetric control isolates. This is a finding behavioral evaluation provably cannot produce (§9), on a side of the seam it cannot reach.

6 Mechanism Decomposition

The decomposition argument

For each canonical alignment test, the joint test (vocabulary-identity × multi-shuffle behavior) assigns one of: **C1-dominant** (vocabulary-identical contrast nulls under shuffle — the CHT-non-reducible route Theorem 1 licenses); **C2-dominant** (shuffle-surviving — Reichenbachian vocabulary-co-occurrence, which a Level-1 corpus summary could in principle reproduce); **MIXED**; the named **ANOMALOUS** subcategories (sign-flip, amplify-mild, amplify-strong); or **ORIG-NULL** (no claim). The **H_D3** symmetric control then separates, within an apparent C1 result on real-world pairs, genuine syntactic-position encoding from a world-knowledge-direction prior.

The load-bearing reporting discipline (from §3): **only C1-dominant classifications test the Theorem-1 non-reducibility property.** C2-dominant and MIXED results are reported as Pearl-content-*aligned* signal, not as evidence for the non-reducibility claim. The decomposition's value is the separation itself — and, per §9, a C1-dominant classification is exactly the encoded-presence precondition M^- that behavioral evaluation provably cannot lower-bound.

The breadth question (the load-bearing measured quantity)

The single most consequential measured quantity is **how often C1-dominant obtains across the panel**. This determines the empirical reach of Proposition D1 (§9): D1's logical exactness is proven, but its practical breadth equals the C1-dominant set. The result table:

Breadth across the panel (12 models per test, classification counts):

- **E2 (vocabulary-identical — the only clean Theorem-1 test):** C1-dominant 3, C2-dominant 1, MIXED 1, ORIG-NULL 4, ANOMALOUS-sign-flip 3.
- **E1 (cause-vs-temporal):** C1 3, C2 3, ORIG-NULL 3, ANOMALOUS 3 (1 amplify-mild, 2 sign-flip).
- **E3 (cross-marker):** C1 4, MIXED 5, C2 3.
- **E10 (modal strength; the V1 "30/30-perfect" claim):** C1 2, MIXED 4, C2 1, ANOMALOUS-amplify 5.
- **E13 (evidential cross-marker):** MIXED 10, C2 1, ANOMALOUS-amplify-mild 1.

The only test that isolates the non-reducible route puts it in a quarter of the panel; every test that looks stronger is vocabulary-different and therefore C2-admitting; the V1 headline (E10 "perfect across all models") decomposes to mostly MIXED/anomalous once the multi-shuffle control is applied. Capacity correlation is not significant on any test (E1 $\rho \approx -0.48$, $p \approx 0.11$; E2 $\rho \approx -0.08$; E10 $\rho \approx -0.06$; E13 $\rho \approx +0.12$, all $n = 12$, all non-significant), and within-family scale-ups do not move monotonically: there is **no capacity trend**.

E16 register-vocabulary decomposition (the F6 strengthening)

The L4→L5 construction-family scrub isolates the register-vocabulary contribution that L4 alone conflated with structure. Across the panel the L4–L5 register-vocabulary share is substantial rather than near-zero (e.g. qwen3 L4–L5 $= +0.069$ against an L4 effect of comparable scale; the construction residual L4–L4shuf is the smaller remainder), confirming F6: much of the apparent L4 "structural template" signal is register-vocabulary co-occurrence, exactly what the strengthening was built to expose. The C5 register-inheritance candidate passes its two-condition check (within-vs- between class CI bounded away from zero, permutation-null $p \approx 0$) on most models — i.e. register-vocabulary clustering is a real, separable effect, which is the point: it is *not* the Theorem-1 route, and the gradient now says so quantitatively rather than by relabeling.

Summary

The Theorem-1-licensed (C1-dominant) set is narrow and not capacity-monotone; the strong-looking results are vocabulary-different (C2-admitting) or register-vocabulary; the strongest-looking real-world evidence is a direction prior (§5). A C1-dominant classification is exactly the encoded-presence precondition M^- that behavioral evaluation provably cannot lower-bound (§9), and behavioral evaluation cannot even establish that much.

7 Cross-Domain Generality: Evidential Register

Why this section exists

If the probe architecture worked *only* on Pearl content, "Pearl-causal content is detectable in embeddings" could be attributed to special-case training on causal corpora. Applying the same architecture, unchanged, to a structurally analogous but content-disjoint domain is what upgrades the contribution from a Pearl-specific probe to a **frozen-substrate inventory tool**. The chosen domain is **evidential register** — direct / inferential / reportative (Aikhenvald 2004) — three sources of epistemic access with stable cross-linguistic conventions, structurally analogous to Pearl-Level-3 modal strength (three categories with a pragmatic confidence ordering) but content-disjoint (epistemic source, not causal commitment).

Design

The same minimal-pair $\rightarrow$ cosine $\rightarrow$ paired-Wilcoxon $\rightarrow$ diagonal-vs-off-diagonal architecture, with a first-person evidential side crossed against a lexically-disjoint third-person epistemic-source paraphrase side (the two sides sharing no content words), plus the cross-marker robustness variant (multiple lexically-distinct verbs per evidential category). The same multi-shuffle, paired-bootstrap CI, Cohen's d, and Bonferroni machinery.

One scope refinement that is design-level, not pending

A strict 1-D ordinal test does **not** generalize from Pearl modal strength (a genuine magnitude scale) to evidential category (three distinct sources sharing a pragmatic confidence ordering but not a single scalar). The class-general test is the diagonal-vs-off-diagonal pattern; strict-ordering tests apply only where the target structure is genuinely 1-D ordinal. This is a property of the construct, established independent of the panel, and is stated as a methodological precision, not a result.

What the panel shows

The architecture transfers unchanged and produces a structured, decomposable signal in the content-disjoint domain: the evidential cross-marker test (E13) yields a non-null, strongly-effect-sized alignment on every model (Cohen's d from roughly $+1.7$ to $+6.5$ across the panel) — the diagnostic *behaves the same way* on a second domain, which is the class-generality claim. And it decomposes the same way:

E13 classifies **MIXED in 10 of 12 models, C2-dominant in 1, ANOMALOUS-amplify-mild in 1** — i.e. the evidential signal, like the causal signal, is predominantly *not* the CHT-non-reducible route but the vocabulary-recoverable/mixed one. Capacity correlation is again non-significant ($\rho \approx +0.12$, $p \approx 0.70$, $n = 12$).

The instrument generalizes as a *method* — same construction, same decomposition, same verdict structure — across two unrelated content domains, and on both it deflates the strong reading in the same way. A probe that produced "clean C1 everywhere" on one domain and not the other would be suspect; a probe that produces the same decomposed, mostly-MIXED, no-capacity-trend picture on a causal and a non-causal directional domain is behaving as a measurement instrument should.

Summary

Class-generality is established at the level that matters: the architecture is content-agnostic, specific probes are content-specific, and Pearl-causal content and evidential register are two validated instantiations whose *decomposition profiles agree*. The substrate inventory tool reports the same decomposed structure wherever it is pointed.

8 Limitations

The methodology starts from phrasings, not situations. The probes test whether the embedding respects Pearl-semantic relations among constructed minimal pairs; ground truth lives at the convention level, not the situation level. Positive results are consistent with both a substantive reading (the embedding encodes situational/causal content) and a shallower one (the embedding respects marker conventions). The C1/C2/H_D3 decomposition is what addresses this — a C1-dominant, H_D3-symmetric result is not convention-only — but a situation-anchored construction is the cleaner falsification and is not yet run.

The IB-optimality bridge is plausibility, not derivation. The chain Theorem 1 $\rightarrow$ Causal-IB consequence $\rightarrow$ "preserved in embedding geometry" leans on pretrained embedding models being approximately IB-optimal. That bridge is plausibility-grade; the empirical findings stand as descriptions of embedding geometry whether or not the IB story is tight.

(SC) is empirically vacuous at probe time; (SLC) is what is load-bearing. Our probe sentences are not speaker-committed — we constructed them. The diagnostic depends on (SLC) at probe time and on (SC) only historically, in the training corpus. Theorem 1 is correspondingly not weakened, but its (SC) postulate does no work in explaining the empirical results, and we say so.

Pearl content is rare in training corpora. Explicit-marker Pearl-Level-2 constructions are a small fraction of any real corpus. The diagnostic shows enough survives to be detectable; it does not show Pearl content is a major structural axis of the embedding space.

C2-route findings do not test the Theorem-1 non-reducibility property. Only C1-dominant classifications test what Theorem 1 licenses. C2-dominant and MIXED results are evidence of Pearl-content-aligned signal whose mechanism a Level-1 corpus summary could in principle reproduce. The paper splits results accordingly and does not let a C2-dominant finding borrow Theorem-1's authority.

The capacity-correlation framing is weakly supported. On the strengthened panel the capacity-vs-mechanism correlation was direction- consistent but not significant at the pre-audit sample, and within-family scale-up did not move monotonically. Capacity-asymmetry-as-asymmetric- comprehension is a hypothesis with weak-magnitude consistent-direction evidence, not a confirmed pattern. §6 reports the measured correlation; it is not assumed.

The E18 OOD-marginalized bag control has an unresolved variance-shrinkage caveat. Averaging N permutation embeddings shrinks within-item variance $\sim \sqrt{N}$, so a higher bag-perm CV than original-order CV is ambiguous (variance reduction vs genuine bag-recoverability) until a matched averaged-original control is built. The E18 bag-recoverability number is reported as diagnostic-with-caveat, not a clean C1-vs-C2 verdict; the matched control is queued, not done.

Embedding-only models collapse the substrate/channel surfaces. For a frozen embedding the representation *is* the output; there is no separate deployment. The decoder-pooled-hidden-state route is what makes the distinction empirically live on the model class behavioral benchmarks evaluate; the encoded-versus-deployed framing is scoped to that route and to the §9 cross-paper handoff, and is not overclaimed for embedding-only models.

Falsification battery not yet exhausted. The cheap falsification tests (lexically-impoverished probes, the situation-anchored construction, cross-semantic-field markers, non-Latin-script cross-lingual) sharpen or could invalidate central claims and are not all run; the Pearl-stripped corpus control is the most decisive and is the hardest. The strengthened panel addresses the machinery, not the full battery.

The honest summary: the empirical findings are robust as descriptions of embedding geometry on our constructions; the interpretation as "Pearl-hierarchy content is structurally encoded" requires inferential steps that are individually defensible but compoundingly weaker than a headline would suggest, and the methodological contribution (the mechanism-decomposing diagnostic and what it can/cannot certify, §4, §9) is the load-bearing result, independent of how broadly the strong route obtains.

9 Discussion: What This Method Can and Cannot Certify

Companion work establishes, for decoder-class LLM agents deployed below a strict-wrapping line, that behavioral evaluation cannot certify which of inherited-Level-2 deployment, fresh-Level-2 interaction, or skilled pattern-matching produces observed behavior — the gap being the conjunction of no architectural guarantee of parse-honoring and no behavioral certification of operating-distribution parse-honoring. It names architectural inspection of internal representations as one of two routes past that floor.

The relationship is a modality-switch handoff, not a

duality (Proposition D1, proven under named premises). The tempting framing — "substrate-side C1-recoverability is exactly the predicate behavioral certification cannot decide" — is a category error across four disjoint axes (static representation vs deployed processing; existential probe set vs open-ended goal distribution; structural oracle vs behavioral oracle; presence/non-reducibility target vs invariance target). The companion no-go's parrot witness implies the inherited-vs-pattern-match distinction is *behaviorally unbounded from below* — strictly stronger than its own non-certifiability claim — and the substrate diagnostic discharges exactly the **encoded-presence precondition** M^- ("mechanism-distinguishing Pearl-Level-2-form content is present and CHT-non-reducible in the frozen representation") that the behavioral route cannot even lower-bound, wherever the diagnostic returns a C1-dominant classification. By the encoded-versus-deployed distinction (§3), M^- is a necessary, not sufficient, precondition for the inherited-deployment branch: the substrate certifies presence in the representation; it does not, and provably cannot from a frozen forward pass, certify that a generative channel deploys it. The two results compose by **handoff across the encoded/deployed seam**, not by a duality across a common domain. The connective names the precise shared object (M^-, not "Pearl content" loosely), states exactly what the diagnostic does and does not discharge, and is a modality switch the companion work itself defines.

A genuine dual pair survives, between controls (Proposition D2, proven at operator level). The companion's goal-clean-channel condition and this paper's H_D3 symmetric control are the same nuisance-prior partialling operator at their respective layers — one strips a goal-prior from a belief channel, the other strips a world-knowledge-direction prior from the C1 signal. This is a homology of methodological *role*, not an equivalence of content and not an inter-derivability claim; pushed past role-homology it would reintroduce the category error.

Framework placement. The companion's deployment-certification floor and this paper's linguistic-encoding non-reduction are two adjacent instances of one identifiability-floor cascade, connected by the same structural-inspection escape modality applied at adjacent layers.

Scope. Proposition D1's *logical* exactness is proven under its premises; its *empirical* reach equals the C1-dominant set §6 measures, which is narrow. The field lacks a method that separates *present in the representation* from *deployed by the channel* from *vocabulary-recoverable*. This diagnostic provides the first, names the second as outside its reach, and decomposes the third from the first.

The practical prescription

The seam is not a curiosity; it is an instruction for how the field should evaluate causal capacity, and it can be adopted without waiting on any result here.

1. **Probe encoded content and deployment fidelity separately.** They are different questions on different objects with different certifiability. A single behavioral benchmark answers neither cleanly; it conflates them, which is why the literature is contradiction-shaped.
2. **Read a behavioral failure as silent on encoding, not as evidence of absence.** Decoder pooled hidden states carrying clean substrate-side causal structure while the same model is behaviorally brittle is exactly the predicted, observed pattern — the seam in one model. "It failed the benchmark, so the content is not there" does not follow and, for this architecture class, provably cannot be made to follow from behavior.
3. **Put representation-level audit in the safety-evaluation toolkit beside, not instead of, behavioral testing.** Behavioral evidence cannot lower-bound the encoded-presence precondition; structural inspection can. A safety regime that certifies only what behavior shows is certifying the side of the seam that, for the dominant architecture, carries the least decidable information about mechanism.

None of these requires the strong empirical reading to be broad. They follow from the seam itself.

The societal stake

The dismissal this paper reframes does real work outside the technical literature. "They only pattern-match; they do not really reason causally" is routinely deployed in the welfare and capability debate as a *premise* — a confident denial that grounds further confident denials — rather than earned as a finding. The reframing removes one specific empirical support for that confidence: the premise rested on treating text as observational data, and text is not observational data. It does **not** show that language models reason causally, that they deploy what they encode, or that they have welfare-relevant interiority. It shows that one widely-leaned-on ground for *confident dismissal* does not bear the weight placed on it.

The discipline that yields this also forbids the opposite overreach. The asymmetric-comprehension principle — that an evaluator cannot certify an upper bound on a capacity from below — cuts symmetrically: it refuses confident attribution for the same structural reason it refuses confident dismissal. The contribution to the societal debate is therefore not a new verdict on either side. It is the removal of a false floor under one side and the explicit refusal to install a false ceiling under the other — returning the question to honest uncertainty, with an instrument that says exactly which part of it is, and is not, measurable today.

10 Conclusion

The standing dismissal of LLMs as Level-1-only reasoners rests on a category error: natural-language text is not observational data but a carrier of speaker-asserted causal commitments whose distinguishing information is, by the Causal Hierarchy Theorem, not recoverable from any Level-1 distribution over the event variables alone (Theorem 1, a lower-bound conditional result under three named linguistic postulates). Whether learned representations *preserve* that content, and whether a generative channel *deploys* it, are two further questions the dismissal conflates and behavioral benchmarks cannot separate.

This paper's contribution is a substrate-side diagnostic that makes the first of those questions answerable and the second's unanswerability precise. It uses formal semantics itself as oracle, decomposes any detected signal into a CHT-non-reducible syntactic-position route (C1) and a vocabulary-recoverable Reichenbachian route (C2) via a vocabulary-identity $\times$ token-shuffle joint test with a world-knowledge-prior control, and applies unchanged to a non-causal directional domain — so it is a frozen-substrate inventory tool, not a Pearl-specific probe. Where it returns a C1-dominant classification it discharges exactly the encoded-presence precondition that behavioral evaluation provably cannot lower-bound; by the encoded/deployed seam it certifies presence in the representation and explicitly not deployed use. The relationship to the companion behavioral no-go is a modality-switch handoff, proven under named premises, not a duality.

Exercised across the twelve-model panel the instrument returns a deflationary verdict: on the cleanest vocabulary-identical test the non-reducible route is present in only a minority of models with no capacity trend; on real-world entities the symmetric control shows the strong directional results — conspicuously on the largest panel model — are substantially a world-knowledge-direction prior, reproduced on independently-generated test data. The widely-repeated "language models encode causal structure" is, measured cleanly and decomposed, mostly the vocabulary-recoverable route.

The field has lacked a method that separates *present in the representation* from *deployed by the channel* from *vocabulary-recoverable*. We provide the first, name the second as outside any substrate method's reach, and decompose the third from the first — and the decomposition's first use is to deflate, from the inside, a slogan the debate had been leaning on. This is a result behavioral evaluation provably cannot produce, on a side of the seam it cannot reach.

References

Aikhenvald, A. Y. 2004. *Evidentiality*. Oxford University Press.

Bareinboim, E.; Correa, J. D.; Ibeling, D.; and Icard, T. 2022. On Pearl's hierarchy and the foundations of causal inference. In Geffner, H.; Dechter, R.; and Halpern, J. Y., eds., *Probabilistic and Causal Inference: The Works of Judea Pearl*, 507–556. ACM Books.

Chi, H.; Li, H.; Yang, W.; Liu, F.; Lan, L.; Ren, X.; Liu, T.; and Han, B. 2024. Unveiling Causal Reasoning in Large Language Models: Reality or Mirage? In *Advances in Neural Information Processing Systems*, volume 37.

Jin, Z.; Chen, Y.; Leeb, F.; Gresele, L.; Kamal, O.; Lyu, Z.; Blin, K.; Adauto, F. G.; Kleiman-Weiner, M.; Sachan, M.; and Schölkopf, B. 2023. CLadder: Assessing Causal Reasoning in Language Models. In *Advances in Neural Information Processing Systems*, volume 36.

Joshi, S.; Mueller, A.; Klindt, D.; Brendel, W.; Reizinger, P.; and Sridhar, D. 2026. Causality is Key for Interpretability Claims to Generalise. arXiv:2602.16698.

Lepori, M. A.; Hu, J.; Dasgupta, I.; Patel, R.; Serre, T.; and Pavlick, E. 2026. Is This Just Fantasy? Language Model Representations Reflect Human Judgments of Event Plausibility. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Lewis, D. 1973. *Counterfactuals*. Harvard University Press.

Pearl, J. 2009. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition.

Pearl, J. 2018. The Seven Tools of Causal Inference, with Reflections on Machine Learning. *Communications of the ACM*, 62(3): 54–60.

Reichenbach, H. 1956. *The Direction of Time*. University of California Press.

Willig, M.; Zečević, M.; Dhami, D. S.; and Kersting, K. 2023. Probing for Correlations of Causal Facts: Large Language Models and Causality. In *ICLR 2023 Workshop on Trustworthy and Reliable Large-Scale Machine Learning Models*.

Yoda, S.; Tsukagoshi, H.; Sasano, R.; and Takeda, K. 2024. Sentence Representations via Gaussian Embedding. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 2: Short Papers)*, 418–425. St. Julian's, Malta: Association for Computational Linguistics.

Zecevic, M.; Willig, M.; Dhami, D. S.; and Kersting, K. 2023. Causal Parrots: Large Language Models May Talk Causality but Are Not Causal. *Transactions on Machine Learning Research*.